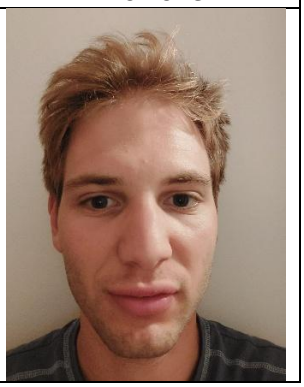
Lab2 – The Metronome Thing – 2026

Group name: Boar
Thing name: The Metronome Thing

Members			
Alessandro Querzé	Andrei Radu	Enrico Giannini	Mirko Tonelli
			

Project Description

In this project we developed a digital metronome Thing using a Raspberry Pi 4B, two push buttons, a green LED and a red LED. The device works in two modes: in play mode the green LED and speaker pulse at the current BPM, while in learn mode the user repeatedly taps a button at the desired tempo. A second button switches the mode, while at least four taps are required to calculate a valid BPM.

We implemented the embedded application in C++ using WiringPi to manage the GPIO pins and cpprestsdk to expose the REST services. The program calculates the BPM from the average interval between consecutive taps and keeps track of the minimum and maximum non-zero BPM values. The endpoints /bpm/, /bpm/min/ and /bpm/max/ allow the current value to be read or changed and the recorded bounds to be read or reset. We also developed a dashboard which uses HTML, CSS and JavaScript that connects to the Raspberry through port 8080, displaying the BPM values and allowing remote control.

Distribution of Effort

Everyone's effort was necessary to understand and implement the overall project. Enrico worked in the design and implementation of the dashboard, Mirko and Alessandro worked on creating and debugging the circuit, Mirko and Andrei provided crucial help in the debugging and hot fixing of the code of the metronome, which at first didn't work as intended.

Relative % of Effort

- Alessandro: 25%
- Andrei: 25%
- Enrico: 25%
- Mirko: 25%